Table 8: Ablation study

Corrected LightSAKT · five-seed test AUC

Architecture	ASSIST2009	ASSIST2015	Algebra 2005–2006
Default	**0.751056 ± 0.000766**	0.726088 ± 0.000315	0.810590 ± 0.000422
No Dropout	0.749340 ± 0.000716	0.725149 ± 0.000593	0.809160 ± 0.000459
No Position Encoding	0.750798 ± 0.000583	0.725749 ± 0.000183	0.811410 ± 0.000603
Single Head	0.750791 ± 0.000765	**0.726622 ± 0.000426**	**0.811755 ± 0.000448**
Without attention	0.708307 ± 0.001107	0.699283 ± 0.000443	0.772550 ± 0.000515
No layer normalization*	*Not run*	*Not run*	*Not run*
Without regularization†	0.749340 ± 0.000716	0.725149 ± 0.000593	0.809160 ± 0.000459

Held-out test AUC: mean ± sample SD over five seeds. Bold marks the highest measured mean in each dataset.

* Not run: the original image includes layer-normalization removal, but no verified experiment exists for it.

† Without regularization equals No Dropout (same evidence); it is not an independent ablation.